

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. Examination (CE)

Dec 2013-Jan 2014

CE720 Object Orientation & Modeling Analytics (OO&MA)

Date: 06.01.2014, Monday Time: 10:00 a.m. To 01:00 p.m. Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Rough work is to be done in the last page of main supplementary, please don't write anything on the question paper.
5. Indicate clearly, the option(s) you attempt along with its respective question no.
6. Figures to the right indicate marks.

SECTION-I

Q-1 Answer the following questions.

1. Briefly discuss the general activities in performing Object Oriented Analysis. 3
2. What is the role of UML (Unified Modeling Language) in framing the architectures for the software? 3
3. What is behavioral modeling? Justify: The purpose of a class diagram is to show the structural makeup of an application. 3
4. The class diagram and interaction diagram should be mutually consistent. Justify. 2

Q-2

- [A] What is the key idea driving object oriented design? How object oriented design provide separation of interface and implementation? 4
- [B] What is your reaction, if you encounter a class with large class size? What is/are the solution(s) for that? 4
- [C] Justify: Only one way to affect an object's state and that is through its operation. Also mention, do we need to consider the other classes while defining and designing the operations of the class. 4

OR

Q-2

- [A] Explain categories of models defined by OMT. Explain rumbough's object modeling technique in detail. 4
- [B] Design a DFD for an institute management system by describing the system with the problem statement. 4
- [C] Discuss how design pattern solve design problems. Give one structured design pattern with a suitable application. 4

Q-3

- [A] Prepare Use-case Diagram, Interaction Diagram, and Activity Diagram for the following System: 6

A customer registers as a member by filling form and mailing it back to the company. No transactions for a period of one-year means he is removed from list and needs to reapply. A member informs the company for any change in personal details. A member makes an order by filling out a sales

order form and then faxing it to the company (alternative is by phone). The assistant always checks the validity of the membership. Order processing clerk checks the availability of the ordered items and hold them for the order. If all items are available, clerk schedules delivery. Inventory control clerk controls and maintains the level of stock and responsible for reordering. For queries member phone the assistant. Members may return the defective goods within 30 days and get their money back. Each task carried out by the system has the name and ID of the staff concerned recorded in to the system.

- [B] Prepare Class Diagram, Component Diagram and Deployment Diagrams for the following System: 6

For a Social Networking Site, consider the functionality like :Stake holders and Interests, Update User Account, Inform Friend about S.N. site, Send Invite to other user, Process Friend Request, Search Friend in List, Send a message to a friend or in group, Update photo Album, Handle Login Failure. Identify the relevant use cases and prepare use-case diagram.

OR

Q-3

- [A] Differentiate: Composition Versus Inheritance. 4
- [B] Explain Component Based Software Development (CBSD) and organization for component management. What are the fundamental issues in developing Real Time Systems? 4
- [C] Justify and give your comments for the following Heuristics in brief: 4
1. A class should capture one and only one key abstraction.
 2. Eliminate irrelevant classes from your diagram.
 3. Minimize the no. of messages sends between classes.
 4. Inheritance should be used only to model a specialization hierarchy.

SECTION-II

Q-4

1. State the phases of unified process model and correlate them with the generic framework activities. 3
2. Describe the problems with modeling behavior in an abstract class? 4
3. What do you mean by Physical Model? Differentiate between component and deployment diagrams. Are they part of Analysis, Design or Implementation? Can we prepare these diagrams at that stage only? What are the additional design & implementation diagrams offered by UML, if any? Define each of them. 4

Q-5

- [A] Which activities are to be performed for the construction of state chart diagram(s)? What is/are state chart diagram(s) used for? How it differs from event trace diagram? 4
- [B] What is the meaning of Use-Case? How to identify use-cases in Use-Case diagram? How are they related with interaction diagram? 4
- [C] What are the relationships between various UML models? Explain in detail about all the UML diagrams that represent only the dynamic aspects of the software system. 4

OR

Q-5

- [A] What is the distinctive purpose of sequence diagram and collaboration diagram? Draw these two diagrams for a scenario at the checkout point at super shop for paying cash for purchase of five items. The items are bar coded. 4
- [B] Explain M-V-C architecture/framework for object oriented programming languages. What are the advantages of it? 4
- [C] What is a UML package diagram? Describe basic symbols and notifications used to show visibility, packages, dependency in Package Diagrams. 4

Q-6

- [A] What is a software design pattern? What is factory pattern? What is builder pattern? What is prototype pattern? 4
- [B] Do you think that testing Object Oriented programs are easier than testing procedural programs? How the object oriented features (inheritance, encapsulation, polymorphism, dynamic binding) influence effective test case design? 4

OR

- [B] Create a state diagram that recognizes an odd number of 0 digit and an even number of 1 digit. 4
- [C] Compare and contrast Object Oriented Programming languages like c++, java & c# with respect to the object oriented concepts. 4

OR

- [C] Prepare the following UML diagrams for Hotel Management System: 4
1. Class
 2. Interaction